

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: CYBER & INFORMATION SECURITY

Practical No. 4

AIM: Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

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CODE:

```
T098 Vikas.py - C:/Users/PRIYANKA/Downloads/T098 Vikas.py (3.13.1)
File Edit Format Run Options Window Help
import hashlib

# RSA keys
p = 61
q = 53
n = p * q
phi = (p - 1) * (q - 1)

e = 17
d = 2753

def sign(message):
    # Create hash
    h = int(hashlib.sha256(message.encode()).hexdigest(), 16)

    # RSA signing
    signature = pow(h, d, n)

    return signature

def verify(message, signature):
    # Create hash again
    h = int(hashlib.sha256(message.encode()).hexdigest(), 16)

    # RSA verification
    decrypted = pow(signature, e, n)

    return decrypted == h % n

print("==== RSA DIGITAL SIGNATURE =====")
message = input("Enter message: ")
signature = sign(message)
print("\nMessage:", message)
print("Digital Signature:", signature)
```

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# Create hash
h = int(hashlib.sha256(message.encode()).hexdigest(), 16)

# RSA signing
signature = pow(h, d, n)

return signature

def verify(message, signature):
    # Create hash again
    h = int(hashlib.sha256(message.encode()).hexdigest(), 16)

    # RSA verification
    decrypted = pow(signature, e, n)

    return decrypted == h % n

print("==== RSA DIGITAL SIGNATURE =====")
message = input("Enter message: ")
signature = sign(message)
print("\nMessage:", message)
print("Digital Signature:", signature)

if verify(message, signature):
    print("Signature Verified: VALID")
else:
    print("Signature Verified: INVALID")

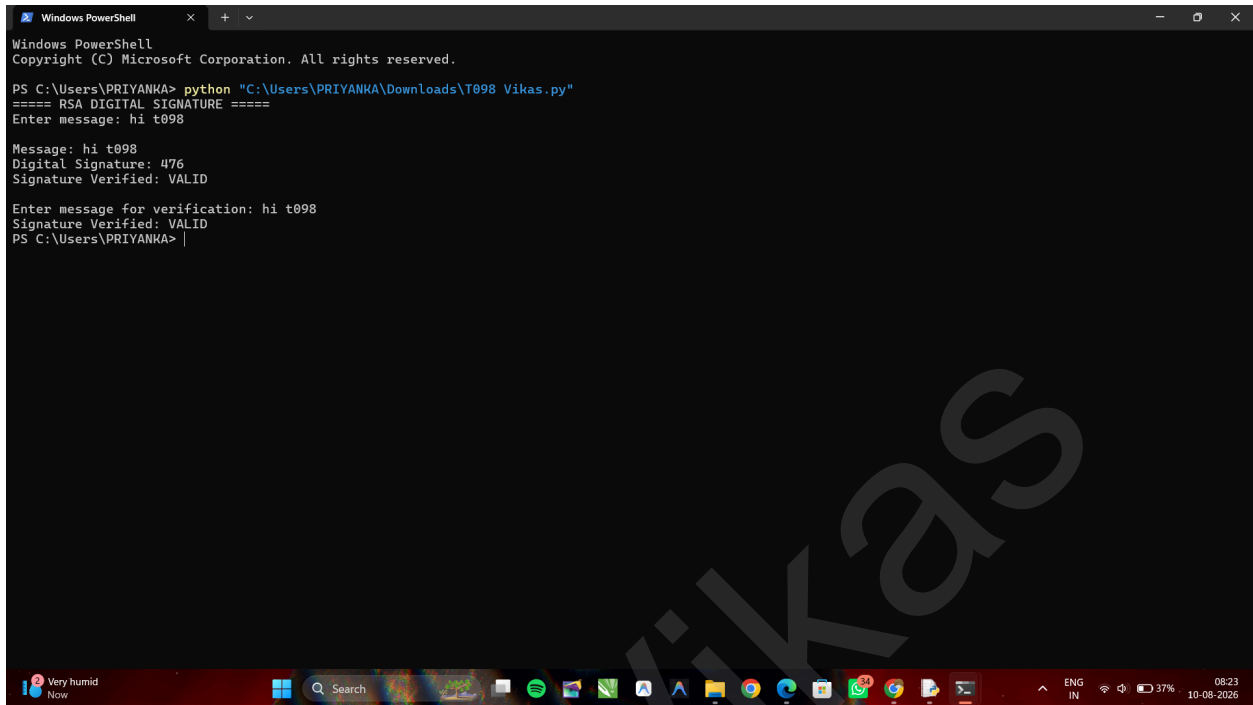
# Test modified message
modified = input("\nEnter message for verification: ")

if verify(modified, signature):
    print("Signature Verified: VALID")
else:
    print("Signature Verified: INVALID")
```

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OUTPUT:



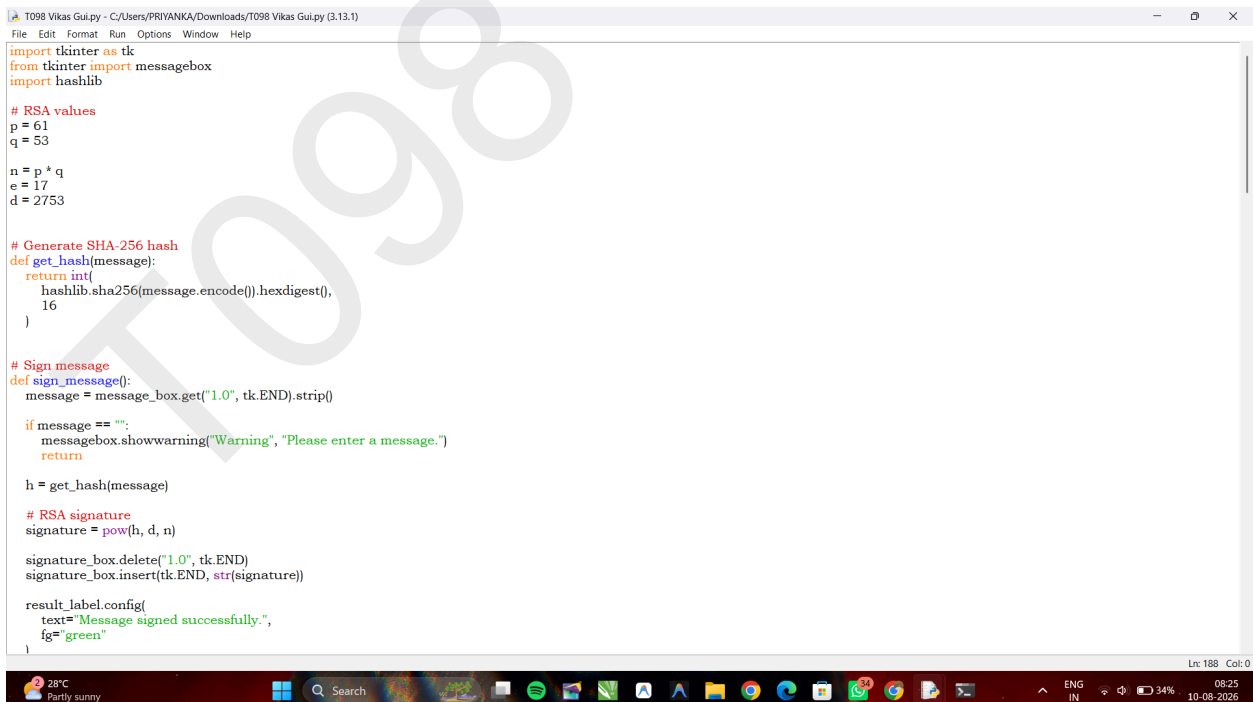
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\PRIYANKA> python "C:\Users\PRIYANKA\Downloads\T098 Vikas.py"
===== RSA DIGITAL SIGNATURE =====
Enter message: hi t098

Message: hi t098
Digital Signature: 476
Signature Verified: VALID

Enter message for verification: hi t098
Signature Verified: VALID
PS C:\Users\PRIYANKA> |
```

CODE GUI:



```
T098 Vikas GUI.py - C:\Users\PRIYANKA\Downloads\T098 Vikas GUI.py (3.13.1)
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import tkinter as tk
from tkinter import messagebox
import hashlib

# RSA values
p = 61
q = 53

n = p * q
e = 17
d = 2753

# Generate SHA-256 hash
def get_hash(message):
    return int(
        hashlib.sha256(message.encode()).hexdigest(),
        16
    )

# Sign message
def sign_message():
    message = message_box.get("1.0", tk.END).strip()

    if message == "":
        messagebox.showwarning("Warning", "Please enter a message.")
        return

    h = get_hash(message)

    # RSA signature
    signature = pow(h, d, n)

    signature_box.delete("1.0", tk.END)
    signature_box.insert(tk.END, str(signature))

    result_label.config(
        text="Message signed successfully.",
        fg="green"
    )
```

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```
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)

# Verify signature
def verify_signature():
    message = message_box.get("1.0", tk.END).strip()
    signature = signature_box.get("1.0", tk.END).strip()

    if message == "":
        messagebox.showwarning("Warning", "Please enter a message.")
        return

    if signature == "":
        messagebox.showwarning(
            "Warning",
            "Please sign the message first."
        )
        return

    try:
        signature = int(signature)

        h = get_hash(message)

        # RSA verification
        verified_hash = pow(signature, e, n)

        if verified_hash == h % n:
            result_label.config(
                text="✓ Signature VALID\nMessage is authentic.",
                fg="green"
            )
        else:
            result_label.config(
                text="✗ Signature INVALID\nMessage has been modified.",
                fg="red"
            )
    except ValueError:
        messagebox.showerror(
            "Error",
            "Invalid signature."
        )

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"Error",
"Invalid signature."
)

# Clear everything
def clear_all():
    message_box.delete("1.0", tk.END)
    signature_box.delete("1.0", tk.END)
    result_label.config(
        text="Result will appear here",
        fg="black"
    )

# ----- GUI -----

root = tk.Tk()
root.title("RSA Digital Signature")
root.geometry("650x500")
root.resizable(False, False)

# Title
tk.Label(
    root,
    text="RSA Digital Signature",
    font=("Arial", 20, "bold")
).pack(pady=20)

# Message
tk.Label(
    root,
    text="Enter Message:",
    font=("Arial", 12, "bold")
).pack(anchor="w", padx=50)

message_box = tk.Text(
    root,
    height=6,
    width=65,
    font=("Arial", 11)

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message_box = tk.Text(
    root,
    height=6,
    width=65,
    font=("Arial", 11)
)
message_box.pack(pady=8)

# Buttons
button_frame = tk.Frame(root)
button_frame.pack(pady=10)

tk.Button(
    button_frame,
    text="Sign Message",
    width=15,
    command=sign_message
).grid(row=0, column=0, padx=5)

tk.Button(
    button_frame,
    text="Verify Signature",
    width=15,
    command=verify_signature
).grid(row=0, column=1, padx=5)

tk.Button(
    button_frame,
    text="Clear",
    width=15,
    command=clear_all
).grid(row=0, column=2, padx=5)

# Signature
tk.Label(
    root,
    text="Digital Signature:",
    font=("Arial", 12, "bold")
).pack(anchor="w", padx=50, pady=(10, 0))

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).grid(row=0, column=2, padx=5)

# Signature
tk.Label(
    root,
    text="Digital Signature:",
    font=("Arial", 12, "bold")
).pack(anchor="w", padx=50, pady=(10, 0))

signature_box = tk.Text(
    root,
    height=4,
    width=65,
    font=("Arial", 10)
)
signature_box.pack(pady=8)

# Result
tk.Label(
    root,
    text="Verification Result:",
    font=("Arial", 12, "bold")
).pack(pady=(10, 5))

result_label = tk.Label(
    root,
    text="Result will appear here",
    font=("Arial", 12)
)
result_label.pack()

# Key information
tk.Label(
    root,
    text="RSA: n = 3233 e = 17 d = 2753 | Hash: SHA-256",
    font=("Arial", 9),
    fg="gray"
).pack(pady=15)

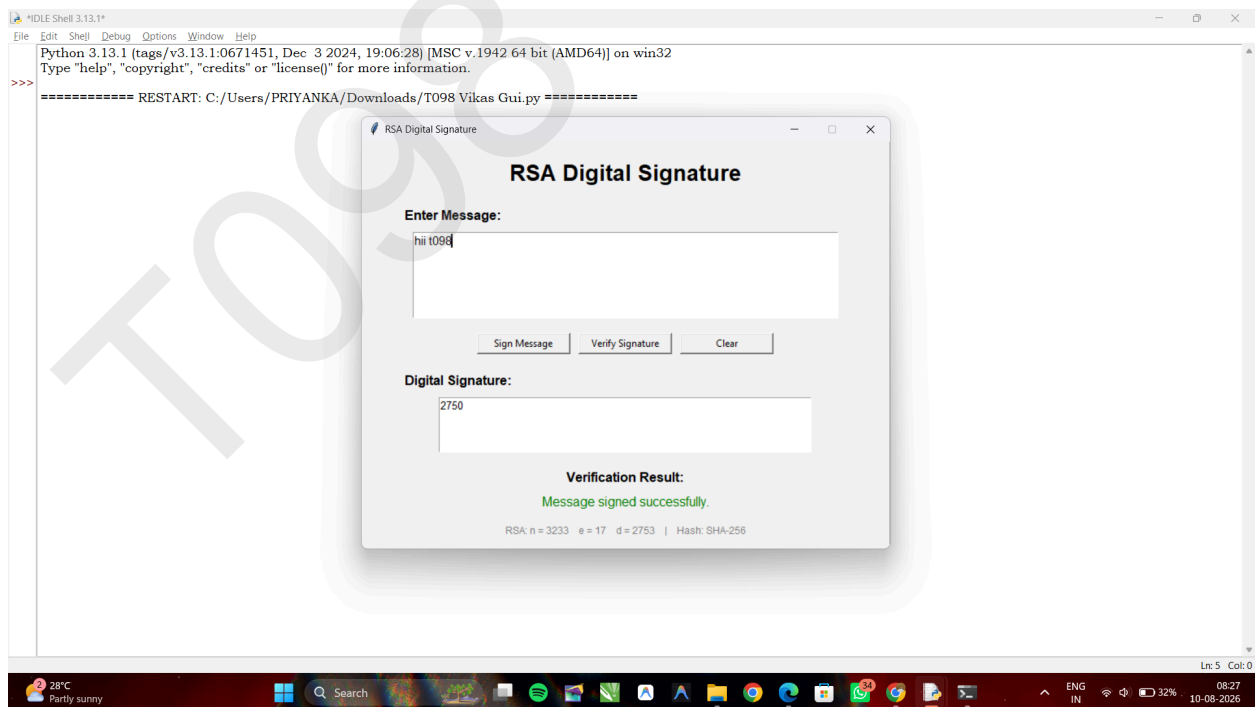
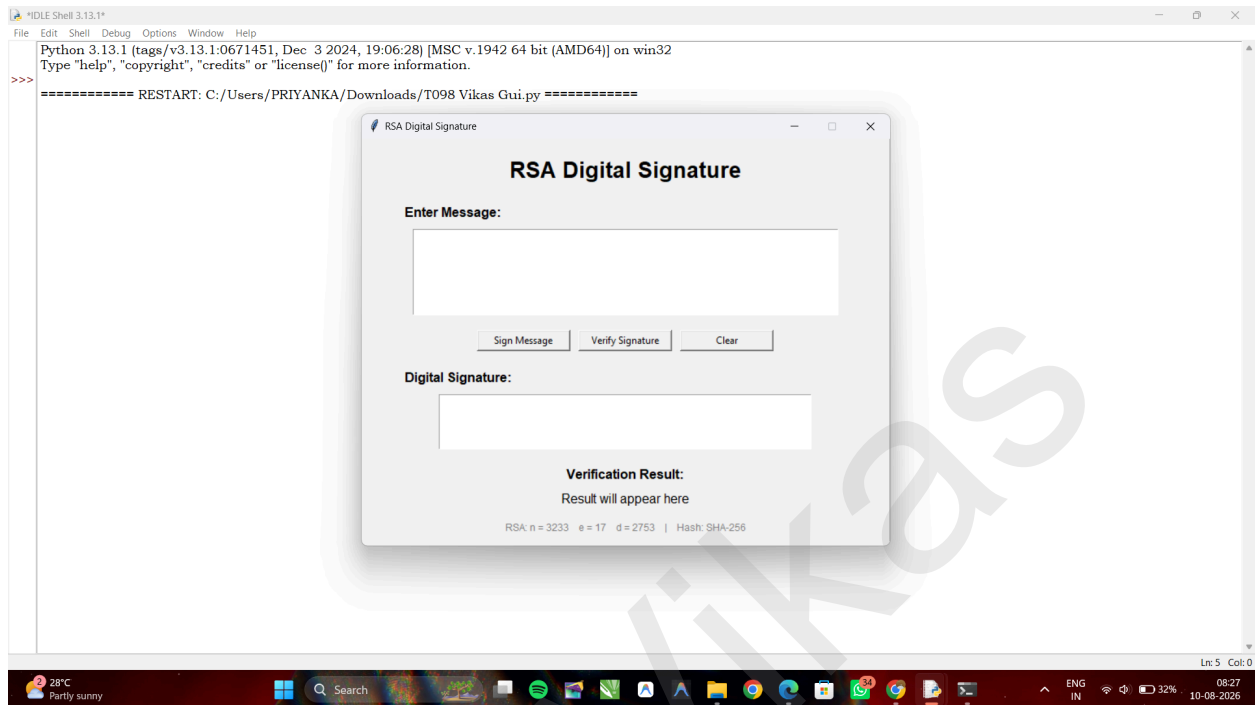
root.mainloop()

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OUTPUT:



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